EcoAI

Problem Statement

The increasing global demand for electricity, combined with climate change and dependence on fossil fuels, has created the need for efficient renewable energy management. Traditional methods of predicting renewable energy generation are often inaccurate due to changing weather conditions and complex environmental factors. There is a need for an intelligent system capable of accurately forecasting renewable energy production while promoting sustainable development.

Project Overview

EcoAI is an AI-powered sustainable energy analytics platform that predicts renewable energy generation using Machine Learning techniques. The project analyzes historical energy and environmental datasets, performs data preprocessing, visualization, feature engineering, and predictive modeling to estimate future renewable energy production. The system enables better energy planning, optimized resource utilization, and supports environmentally sustainable decision-making

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Solution Offered

The proposed solution integrates data preprocessing, exploratory data analysis, machine learning model training, prediction, and performance evaluation into a single workflow. The model identifies important environmental factors affecting renewable energy generation and provides accurate predictions to assist energy providers, researchers, and policymakers.

Who Are The End Users?

- Renewable Energy Companies
- Government Energy Departments
- Environmental Researchers
- Smart Grid Operators

Technology Used To Solve The Problem

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Machine Learning Algorithms